

An Empirical Study of Inadequate and Adequate Test Suite Reduction Approaches

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ABSTRACT

Background. Regression testing is conducted after changes are made to a system in order to ensure that these changes did not alter its expected behavior. The problem with regression testing is that it can require too much time and/or too many resources. This is why researchers have defined a number of regression testing approaches. Among these, Test Suite Reduction (TSR) approaches reduce the size of the original test suites, while preserving their capability to detect faults. TSR approaches can be classified as adequate or inadequate. Adequate approaches reduce test suites so that they completely preserve the test requirements (e.g., statement coverage) of the original test suite, while inadequate ones produce reduced test suites that partially preserve these test requirements.

Aims. We studied adequate and inadequate TSR approaches in terms of tradeoff between reduction in test suite size and loss in fault detection capability. We also considered three different kinds of test requirements (i.e., statement, method, and class coverages). **Method.** We conducted an experiment with six adequate (e.g., HGS) and 12 inadequate (e.g., the inadequate version of HGS) TSR approaches. In this experiment, we considered 19 experimental objects from a public dataset, i.e., SIR (Software-artifact Infrastructure Repository).

Results. The most important result from our experiment is that inadequate approaches, as compared with adequate ones, allow achieving a better tradeoff between reduction in test suite size and loss in fault detection capability. This is especially true when these approaches are applied by considering statement and method coverages as test requirements.

Conclusions. Although our results are not definitive, they might help the tester to choose both TSR approach and kind of code coverage that is closer to her needs when testing a software system.

CCS CONCEPTS

• **Software and its engineering** → **Software testing and debugging**;

KEYWORDS

Regression testing, test suite reduction, controlled experiment

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1 INTRODUCTION

Regression testing is conducted after changes are made to the code of a system to ensure that these changes did not alter its behavior. The problem with regression testing is that it can require too much time and/or too many resources [23]. This is why researchers have defined a number of regression testing approaches, which can be classified into three main categories: (i) Test Suite Reduction (TSR); (ii) regression test case selection; and (iii) test case prioritization [23]. Both TSR and regression test case selection approaches reduce the size of test suites, while preserving their capability to detect faults. TSR approaches delete obsolete or redundant test cases from test suites [14], while regression test case selection approaches select subsets of test cases that exercise changed parts of SUTs (System Under Test) or parts affected by changes [12]. TSR is sometimes called test suite minimization when the deletion of test cases is not permanent. However, these two concepts are used in an interchangeable way [23]. Finally, test case prioritization approaches sort test cases on the basis of some criteria [10].

TSR approaches can be either *adequate* or *inadequate* [15]. Adequate approaches reduce test suites so that they completely satisfy the test requirements of the original test suites. For instance, let statement coverage be the kind of test requirement considered, an adequate TSR approach produces a reduced test suite that covers the same statements as the original test suite. TSR approaches are inadequate (or also non-adequate from here onwards) when the reduced test suites only partially preserve the test requirements of the original test suites.

In this paper, we present an experiment to study adequate and inadequate TSR approaches in terms of tradeoff between reduction in test suite size and loss in fault detection capability. The experiment was conducted on six adequate and 12 inadequate TSR approaches. We also considered three different kinds of test requirements (i.e., statement,¹ method,² and class³ coverages), when reducing test suites. Our experiment is founded on SIR (Software-artifact Infrastructure Repository) [5], a public dataset widely used in the regression testing field (e.g., [16, 24]). We conducted our experiment on 19 experimental objects from this dataset.

¹A statement is covered if it exists at least one test case that executes that statement.

²A method is covered if at least one of its statements is covered.

³A class is covered if at least one of its statements is covered.

Summarizing, we make the following contributions:

- An experiment where we assessed a large number of TSR approaches (18 in total).
- When reducing test suites, we studied three different kinds of test requirements: statement, method, and class coverages.
- Comparisons within adequate approaches, within inadequate approaches, and between them.

Paper Structure. In Section 2, we provide background information. In particular, we report how the TSR problem has been defined in the literature and introduce the adequate/inadequate approaches considered in our experiment. The planning and the design of this experiment are described in Section 3, while the obtained results are presented and discussed in Section 4. Related work is highlighted in Section 5. Final remarks conclude the paper in Section 6.

2 BACKGROUND

Rothermel *et al.* [14] formally defined TSR as follows:

Given: A test suite, T , a set of test requirements $r_1, \dots, r_n$, which must be satisfied to provide the desired adequate testing of the SUT, and subsets of T , $T_1, \dots, T_n$, one associated with each of r_i s such that any one of the test cases t_j belonging to T_i can be used to achieve the requirement r_i .

Problem: Find a representative set, T' , of test cases from T that satisfies all r_i s.

A number of TSR approaches have been proposed in the literature (e.g., [1, 7, 9, 19]). Traditional approaches are based on the aforementioned definition and aim at reducing the size of test suites satisfying all the test requirements. Such a kind of approaches is known as adequate [15]. Researchers have also proposed TSR approaches that relax the constraint of satisfying all the test requirements and have named these approaches as inadequate [15].

In the following subsections, we summarize the adequate and inadequate TSR approaches studied in our experiment. We chose these approaches because they are quite common for experiments in the context of TSR (e.g., [15, 18]). In addition, we considered the inadequate Clustering-Based (CB) approach by Coviello *et al.* [4] since it is a very recent contribution in the TSR field and then it might be considered one of the state-of-the-art approaches.

2.1 Adequate Approaches

2.1.1 GRD (Greedy). This approach is based on a greedy algorithm that Li *et al.* [9] proposed to prioritize test cases. Then, it was applied to reduce test suites (e.g., [18]). GRD selects each time a test case, among those not yet selected, that satisfies the maximum number of unsatisfied test requirements. The selection process concludes when the selected test cases satisfy all the test requirements. The reduced test suite consists of the selected test cases.

2.1.2 HGS (Harrold Gupta Soffa). It is an heuristic-based approach proposed by Harrold *et al.* [7]. HGS groups test requirements into k sets so that the i -th set contains only test requirements satisfied by i test cases. For example, the first set contains test requirements satisfied by only one test case. HGS first selects the test cases satisfying the first set (i.e., $i = 1$) of test requirements. Then, it selects the test cases satisfying the second set (i.e., $i = 2$) and so on. The reduced test suite consists of the selected test cases.

2.1.3 2OPT (2-Optimal). Similar to GRD, this approach was first studied to prioritize test cases (i.e., [9]) and then adapted to reduce test suites (e.g., [18]). 2OPT relies on the 2-Optimal algorithm, which represents an instance of the k -Optimal algorithm when $k = 2$. At each step, 2OPT selects the next two test cases that together satisfy the largest number of test requirements. The selected test cases comprise the reduced test suite.

2.1.4 DGR (Delayed Greedy). This approach was proposed by Tallam and Gupta [19]. DGR postpones the application of the greedy heuristic after executing the following steps: (i) if a set of test requirements satisfied by a test case t_i is a super-set of a set of requirements satisfied by another test case t_j , then t_j is removed from the test suite; and (ii) if a set of test cases satisfying a requirement r_i is a subset of a set of test cases satisfying r_j , then r_i is removed from the test suite.

2.1.5 GE. Chen and Lau [1] proposed this heuristic-based approach. GE first selects all essential test cases—if a test requirement is satisfied by only one test case, this test case is essential—, then it uses a greedy algorithm for the remaining test requirements.

2.1.6 GRE. Chen and Lau [1] proposed an extension of GE that they named GRE. This approach first removes all redundant test cases from a test suite—a redundant test case satisfies only a subset of the test requirements another test case satisfies—, then it applies GE on the remaining test cases.

2.2 Inadequate Approaches

2.2.1 Inadequate Versions of Adequate Approaches. According to Shi *et al.*'s study [15], adequate approaches (e.g., those previously summarized) can be converted into inadequate approaches. Let A be a traditional adequate approach and let $0 \leq l < 100$ be the percentage of test requirements that must be satisfied. The inadequate version of A , i.e., A_l , will produce a reduced test suite, T' , satisfying $l\%$ of the test requirements that the original test suite (i.e., T) satisfied. For example, GRD_l is the inadequate version of GRD satisfying $l\%$ of the test requirements.

2.2.2 Inadequate Clustering-Based (CB) Approach. Coviello *et al.* [4] presented an inadequate CB approach and a number of instances of the process underlying this approach. The reduction of the CB approach goes through the following steps:

- (1) The dissimilarity between each pairs of test cases is computed. Two test cases are similar if they cover nearly the same test requirements; on the contrary, they are dissimilar. To compute the dissimilarity of test cases, several measures can be used. The authors explored the following ones:
 - Euclidean Distance (EUCL);
 - Cosine (Dis)similarity (COS);
 - Jaccard-Based Dissimilarity (JACC);
 - Hamming Distance (HAMM);
 - Levenshtein Distance (LEV);
 - String Kernels-Based Dissimilarity (SK).
- (2) This phase groups test cases that are similar (and then considered redundant) into the same cluster. To this end, a hierarchical agglomerative clustering algorithm is employed. This algorithm returns a hierarchical structure of clusters known

as dendrogram. To produce a set of clusters, the dendrogram has to be cut at a given level. For example, the dendrogram can be cut so as to satisfy 1% of the test requirements.

- (3) For each cluster, the representative test case—the test case covering the highest number of test requirements in the cluster—is selected. The set of representative test cases constitutes the reduced test suite.

The CB approach can be instantiated in different ways, where each instance corresponds to the use of a different dissimilarity measure. For example, CB_{EUC} represents the instance of CB approach where the Euclidean Distance is used to compute the dissimilarity of test cases. The approach by Coviello *et al.* [4] represents the state of the art for inadequate TSR. This is why we studied such approach and its instances in our experiment.

3 EMPIRICAL ASSESSMENT

In this section, we present our experiment. The template used to present the experiment planning is based on that proposed by Wohlin *et al.* [20].

3.1 Goal Definition

This experiment aimed at investigating the following Research Questions (RQs):

- RQ1.** Which kind/s of code coverage (test requirement) and which adequate TSR approach/es (if any) allow achieving the best tradeoff between reduction in test suite size and loss in fault detection capability?
- RQ2.** Which kind/s of code coverage (test requirement) and which inadequate TSR approach/es (if any) allow achieving the best tradeoff between reduction in test suite size and loss in fault detection capability?
- RQ3.** Do inadequate TSR approaches outperform adequate ones in terms of tradeoff between test suite size and loss in fault detection capability?

A TSR approach achieves a good tradeoff between reduction in test suite size and loss in fault detection capability if the reduced tests suites are much smaller than the original ones and detect nearly the same faults as the original test suites.

3.2 Context Selection

The experiment was run in the context of Java software systems. We selected these systems from SIR,⁴ a repository of software-related artifacts whose goal is to support rigorous controlled experiments on regression testing [5]. SIR allows researchers to have information on the faults software systems contain. This information is needed when estimating the loss in fault detection capability of TSR approaches. SIR has been largely used to carry out experiments in the regression testing field (*e.g.*, [16, 24]).

The experimental objects were 19: eight versions for *Ant*, three versions for *JTopas*, five versions for *JMeter*, and three versions for *XMLSecurity*. The experimental objects are the same as Zhang *et al.* used in their investigation [24]. In Table 1, we report, for each experimental object, LOC (Lines of Code), number of (JUnit) test

Table 1: Some information on the experimental objects.

System	Version	LOC	#Test cases	#Faults
Ant	v1 (1.3)	23,796	133	3
	v2 (1.4)	37,478	212	1
	v3 (1.4.1)	37,554	217	2
	v4 (1.5)	64,445	533	8
	v5 (1.5.2)	66,085	539	7
	v6 (1.5.3-1)	66,144	570	1
	v7 (1.6beta)	88,414	842	10
	v8 (1.6beta2)	88,449	845	2
JTopas	v1 (0.4)	4,276	26	5
	v2 (0.5.1)	4,520	28	4
	v3 (0.6)	10,117	56	5
JMeter	v1 (1.8)	33,620	51	4
	v2 (1.8.1)	33,290	63	3
	v3 (1.9.RC1)	37,474	78	8
	v4 (1.9.RC2)	38,613	78	2
	v5 (1.9)	40,989	91	2
XMLSecurity	v1 (1.0.4)	21,601	94	5
	v2 (1.0.5D2)	27,990	94	6
	v3 (1.0.71)	19,731	84	3

cases, and number of faults. It is worth mentioning that the test cases and the faults are those available on SIR.

3.3 Variable Selection

In our experiment, the main independent variable (also known as main factor) is **Approach**. This variable is nominal and assumes values indicating the studied TSR approaches (*e.g.*, *GRD* or *GRD_I* or *CB_{EUC}*). We also took into **Coverage** as independent variable. It is nominal variable, which assumes three possible values: *Statement*, *Method*, and *Class*. For example, *Statement* means that the statement coverage is the test requirement used by a given approach to reduce the test suites.

The TSR approaches studied in this experiment have been assessed on the basis of the reduced test suites they produced. Two evaluation metrics are usually employed in the TSR field [23] and this was why we opted for them in this experiment. These metrics are: Reduction in test suite Size (**RS**) and Reduction in Fault detection capability (**RF**). Let $|T|$ and $|T'|$ be the size of the original and reduced test suites, respectively. Let F be the number of faults the original test suite $|T|$ detects, while let F' be the number of faults detected by the reduced test suite, $|T'|$. RS and RF are defined as:

$$RS = \frac{|T| - |T'|}{|T|} * 100 \quad (1)$$

$$RF = \frac{F - F'}{F} * 100 \quad (2)$$

Both RS and RF assume values in between 0% and 100%. RS reflects the amount of test cases within the original test suite that are not included in the reduced one. A value equal to 0% means that the size of the original test suite has not been reduced. Thus, a high value for RS is desirable. On the other hand, RF indicates the percentage of faults the reduced test suite does not detect with respect to the

⁴sir.unl.edu

faults the original test suite detects. An RF value equal to 0% means that the reduced test suite has preserved the same capability of detecting faults as the original test suite. Thus, a value equal to 0% is desirable. A TSR approach achieves a good tradeoff if the RS values are high while the RF values are low. RS and RF represent the two dependent variables we studied in our experiment.

3.4 Procedure

The experiment consists in applying adequate and inadequate approaches on each experimental object so as to produce reduced test suites. The approaches are those described in Section 2.1 and Section 2.2. Each of them was applied by considering statement, method, and class coverages as test requirements.

For any inadequate approach, the value of l was fixed to 95%. This means that whatever the inadequate approach is (e.g., GRD_I or CB_{EUCI}), it produces reduced test suites satisfying 95% of the test requirements (e.g., GRD_I with statement coverage produces reduced test suites covering 95% of the statements that the original test suites covered). Such a value of l was chosen on the basis of the results Shi *et al.* [15] reported.

Given the reduced test suites, we computed the values for the considered dependent variables: RS and RF. This was possible because we knew both the dimension of the original test suites and the faults contained in the experimental objects.

3.5 Data Analysis

We computed, for any approach and any kind of coverage, some descriptive statistics, which summarize the distributions of the RS and RF values. Then, we built boxplots to graphically represent these distributions. When needed, we applied statistical inference to better support the discussion of the results. In particular, we used the Friedman test [6] to test the effect of Approach and Coverage on the dependent variables. Before applying the Friedman test, we verified if the assumptions to apply linear mixed model analysis methods were verified. Since these assumptions were not verified (neither by applying data transformations), we opted for a non-parametric alternative of these methods, namely the Friedman test (since samples are not independent). We applied this test considering as independent variable the combination of Approach and Coverage and as dependent variables RS and RF. This implies that our statistical inference did not allow us to individually study the effect of Approach and Coverage on the dependent variables.⁵

The reader can find a replication package on the web,⁶ where she can have access to the raw data and the scripts used for the data analysis together with the results of additional analyses not reported in the paper for space reason.

For each statistical test, we decided to accept (as customary) a probability of 5% of committing Type-I-error (i.e., $\alpha = 0.05$).

⁵We also conducted a post-hoc analysis by performing a pairwise comparison between each pair of approaches on both RS and RF. To this end, we used the Wilcoxon signed-rank test. Since we performed multiple comparisons, we used the Benjamini-Hochberg method to adjust the p-values. We did not report this further analysis in the paper for space reason.

⁶www2.unibas.it/sromano/TSR

3.6 Instrumentation

To carry out the experiment, we exploited RAISE, a prototype of a supporting tool, which implements HGS, GRD, 2OPT, and DGR. RAISE is available on the web⁷ and has been used in previous empirical studies on TSR approaches (e.g., [17, 18]). We implemented the six instances of the CB approach in a prototype of a supporting tool written in Java we named CUTER [3]. We also developed a prototype implementing the other adequate (i.e., GE and GRE) and inadequate TSR approaches (e.g., HGS_I).

The coverage information (e.g., statement coverage) of the experimental objects were gathered through JaCoCo.⁸ Then, this information was used by any TSR approach to produce reduced test suites.

3.7 Threats to Validity

To understand strengths and limitations of our experiment, we describe possible threats that might affect the validity of the results.

- Threats to *construct validity* concern the relationship between theory and observation. In our study, this kind of threats is due to the metrics and dataset used. However, the used metrics are widely adopted to quantify the investigated constructs and represent the standard for assessing TSR approaches [23]. As for the dataset, it has been created by external researchers [5] and used in a number of studies on regression testing and TSR, in particular (e.g., [16, 24]). To further mitigate this kind of threat, we plan to replicate our study on different datasets (e.g., Defects4J [8]).
- Threats to *internal validity* concern factors internal to the study that could affect the obtained results. For example, the implementation of the studied TSR approaches might threaten the validity of the results. We partially dealt with this kind of threat by using tools that external researchers implemented and applied in previous empirical studies. Finally, we tested our tools and those the external researchers implemented.
- Threats to *conclusion validity* concern the relationship between the dependent and independent variables. The performed statistical analyses might threaten this kind of validity. In this respect, we applied robust and sensitive statistical tests. Another threat is related to the studied TSR approaches. To deal with this kind of threat, we chose TSR approaches that can be considered a natural choice when performing studies like ours (e.g., [15, 18, 24]).
- Threats to *external validity* concern the generalizability of the results. Although in our investigation we considered software systems previously used in other studies and these systems cover different application domains, we cannot guarantee that our findings can be generalized to the universe of Java programs. Future work is needed to increase the external validity of our study.
- Threats to *reliability validity* concern the possibility of replicating our study. We provide all the details needed to replicate our experiment. Moreover, we made available on the web our replication package comprising raw data, scripts, etc.

⁷code.google.com/archive/p/raise

⁸jacoco.org

Table 2: Some descriptive statistics for RS and RF for each adequate approach and each kind of coverage.

DV	Coverage	Statistic	GRD	HGS	2OPT	DGR	GE	GRE
RS	Statement	median	0	0	0	0	0	0
		mean	1.4	1.42	1.32	1.42	1.4	0.76
		SD	3.15	3.15	2.92	3.15	3.15	1.34
		max	13.19	13.19	12.09	13.19	13.19	4.32
		min	0	0	0	0	0	0
	Method	median	0	0	0	0	0	0
		mean	1.53	1.53	1.43	1.53	1.53	0.89
		SD	3.25	3.25	3.02	3.25	3.25	1.6
		max	13.19	13.19	12.09	13.19	13.19	5.25
		min	0	0	0	0	0	0
	Class	median	74.53	74.06	74.44	74.53	74.53	20.83
		mean	70.56	70.57	69.88	70.64	70.56	23.66
		SD	9.51	9.54	9.79	9.58	9.51	5.1
		max	79.79	79.79	79.65	79.82	79.79	32.14
		min	55.13	55.13	53.85	55.13	55.13	17.65
RF	Statement	median	0	0	0	0	0	0
		mean	0	0	0	0	0	0
		SD	0	0	0	0	0	0
		max	0	0	0	0	0	0
		min	0	0	0	0	0	0
	Method	median	0	0	0	0	0	0
		mean	0	0	0	0	0	0
		SD	0	0	0	0	0	0
		max	0	0	0	0	0	0
		min	0	0	0	0	0	0
	Class	median	25	25	25	20	25	0
		mean	33.72	32.66	32.66	26.38	33.72	13.47
		SD	34.68	33.48	33.48	29.01	34.68	23
		max	100	100	100	100	100	66.67
		min	0	0	0	0	0	0

The dependent variable values are expressed as percentage.

4 RESULTS AND DISCUSSION

In this section, we present and discuss the results according to the defined RQs. We conclude the section by discussing possible implications from the researcher and the practitioner perspectives.

4.1 Investigating RQ1

4.1.1 Results. In Table 2, we provide, for each adequate approach and each kind of coverage, the values of mean, median, Standard Deviation (SD), minimum (min), and maximum (max), for the Dependent Variables (DVs) RS and RF. The distributions of the RS values are graphically summarized by means of the boxplots in Figure 1.a. Similarly, we summarize the distributions of the RF values in Figure 1.b.

The descriptive statistics (see Table 2) and the boxplots (see Figure 1) suggest that the adequate approaches achieve similar values for both RS and RF in case of statement coverage. The RS values are close to 0% with some exceptions (see the outliers in the boxplots), while the RF values are always equal to 0%. A similar trend can be observed for method coverage. As for class coverage, it seems that there are not huge differences among the RS values achieved by the adequate approaches except for GRE, which achieves worse values for RS (e.g., see the boxplots in Figure 1.a). However, GRE exhibits values for RF better than the others (e.g., the median value

for GRE is 0%, while for the other approaches it is equal to either 20% or 25%).

By comparing the kinds of coverage, we can observe that there is not a huge difference when using statement or method coverage. In both the cases, the mean RS values are close to 0%, while the RF values are equal to 0%. The use of class coverage allows any adequate approach to achieve better RS values—e.g., the mean values vary from 23.66% (GRE) to 70.64% (DGR)—at the price of worse RF values—e.g., the mean values ranges in between 13.47% (GRE) and 33.72% (GRD and GE).

We executed a Friedman test on the gathered data and the results suggest the combination of Approach and Coverage is statistically significant on the dependent variables RS and RF (the returned p-values are both equal to $2.2e-16$). That is, there are pairs of adequate approaches and kinds of coverage that are statistically better/worse than others.

4.1.2 Discussion. The results from the exploratory analysis and statistical inference did not allow us to definitively answer RQ1 (which kind/s of code coverage and which adequate TSR approach/es allow achieving the best tradeoff between reduction in test suite size and loss in fault detection capability?). Indeed, when considering statement or method coverage, the adequate approaches perform similarly in terms of tradeoff between reduction in test suite size and loss in fault detection capability. On the other hand, GRE seems to obtain a better tradeoff between reduction in test suite size and loss in fault detection capability, when working with class coverage (this result was also confirmed by our post-hoc analysis). As final recommendation, the obtained results suggest to use GRE with class coverage in case the reduction in test suite size is desirable and the loss in fault detection capability is not a major issue. In case the loss in fault detection capability is an issue, the tester should opt for any of the adequate approaches working with method and statement coverages being conscious that a small reduction in test suite size might be obtained.

4.2 Investigating RQ2

4.2.1 Results. For each inadequate approach and each kind of coverage, we report in Table 3 the values of mean, median, SD, min, and max, for RS and RF. The boxplots for these variables are shown in Figure 2.

The boxplots (see Figure 2.a) suggest that there are differences in the RS values among the inadequate approaches in case of statement coverage: the boxplots for HGS_I and DGR_I are much lower than the others. As shown in Table 3, HGS_I and DGR_I achieve on average RS values equal to 14.26% and 15.48%, respectively. Any other inadequate approach exhibits, on average, better RS values—the best mean value is achieved by CB_{CoS} (30.6%). The boxplots for CB_{CoS} is also comparatively short (see Figure 2.a), this suggests small variations in the RS values (see also the deviation standard in Table 3). With respect to RF, we can observe that the RF values for 2OPT_I and GE_I are always equal to 0%, while they are close to 0% for the other inadequate approaches (see Figure 2.b). On average, RF values are very close to 0% for any inadequate approach.

As for method coverage, the boxplot depicting the RS values for CB_{SK} seem to be lower than others, so suggesting differences in the RS values. The other boxplots for RS do not show huge differences.

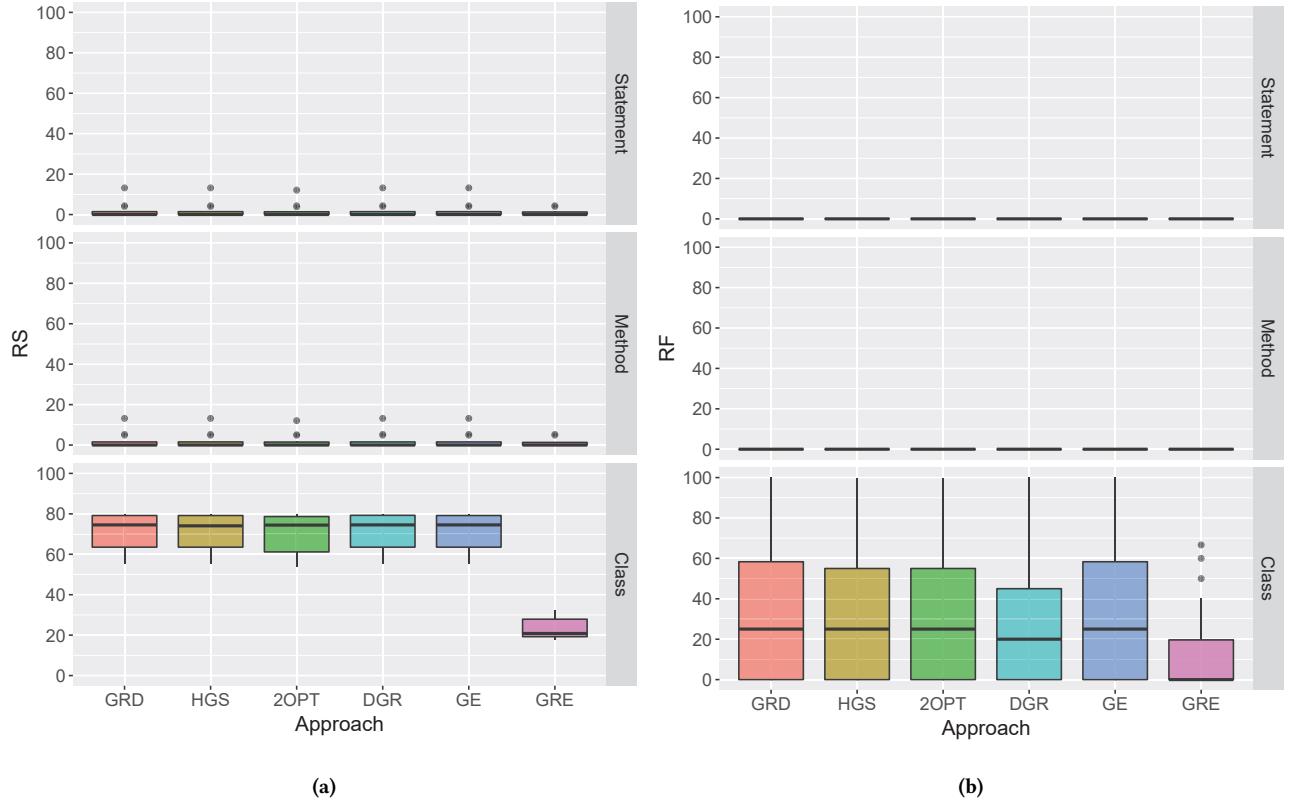


Figure 1: Boxplots for RS (a) and RF (b) for each adequate approach and each kind of coverage.

We can also observe that, on average, CB_{JACC} achieve the best values for RS (29.77%) followed by CB_{COS} (29.13%)—it has exhibited the best value in case of statement coverage. Both CB_{JACC} and CB_{COS} seem to outperform, on average, the other inadequate approaches with respect to RF: the mean values for CB_{JACC} and CB_{COS} are equals to 0.66%, while those for the other inadequate approaches range from 4.04% (CB_{EUC} and CB_{HAMM}) to 13.33% (HGS_I and DGR_I).

As for class coverage, the boxplots for RS and RF (see Figure 2) suggest that, GRD, HGS, 2OPT, DGR, and GE achieve value for RS better than those of the other inadequate approaches, but at the price of worse RF values. Indeed, the instances of the CB approach seem to achieve a better tradeoff between RS and RF, because they obtain high RS values—the mean values range from 41.69% (CB_{SK}) to 44.83% (CB_{LEV})—with low values of RF—the mean values range from 2.07% (CB_{LEV}) to 5.92% (CB_{SK}).

The results of the Friedman test suggest that the combination of Approach and Coverage is statistically significant on the both RS and RF (the returned p-values are both equal to $2.2e-16$). That is, there are pairs of inadequate approaches and kinds of coverage that are statistically better/worse than others.

4.2.2 Discussion. On the basis of the results from the exploratory analysis and statistical inference, it seems that *the instances of the CB approach allow achieving the best tradeoff (as compared with the other inadequate approaches) between reduction in test suite size*

and loss in fault detection capability, when working at class coverage level. In fact, the instances of the CB approach reduce more than the other approaches with a very small effect on the fault detection capability of the reduced test suites (this is confirmed by the results of our post-hoc analysis).

4.3 Investigating RQ3

4.3.1 Results. By looking at Figure 1.a and Figure 2.a, we can observe that, for each kind of coverage, the inadequate approaches achieve RS values higher than those achieved by the adequate ones. For example, when considering statement coverage, the adequate approaches achieves mean RS values ranging from 0.76% (GRE) to 1.42% (HGS and DGR), while the mean RS values for the inadequate approaches range from 14.26% (HGS_I) to 30.6% (CB_{COS}). As for RF, the values for the inadequate approaches are always equal to 0% for statement and method coverages, while most of the RF values of the inadequate approaches are equal to 0% for the same kinds of coverage (see the boxplots in Figure 2.b). When considering class coverage, the adequate approaches achieve high values of RS, but at the price of high values of RF. On the other hand, some inadequate approaches (*i.e.*, CB_{EUC} , CB_{COS} , CB_{JACC} , CB_{HAMM} , CB_{LEV} , CB_{SK}) obtain high values of RS with low values of RF.

Table 3: Some descriptive statistics for RS and RF for each inadequate approach.

DV	Coverage	Statistic	GRD _I	HGS _I	2OPT _I	DGR _I	GE _I	GRE _I	CB _{EUCL}	CB _{COS}	CB _{JACC}	CB _{HAMM}	CB _{LEV}	CB _{SK}
RS	Statement	median	26.37	11.93	25.27	14.29	26.37	24.47	29.5	28.39	28.62	28.71	25.6	23.01
		mean	29.57	14.26	29.66	15.48	29.17	28.98	27.99	30.6	29.26	29.34	23.73	22.7
		SD	13.52	7.78	13.25	8.18	13.82	13.94	8.12	7	8.08	7.48	9.05	10.44
		max	50.69	29.79	50.23	30.85	50.69	50.69	38.3	46.27	46.39	46.39	41.49	46.39
		min	12.7	4.76	11.11	6.06	12.7	12.7	6.04	19.61	14.29	15.38	3.85	7.14
	Method	median	25.68	26.14	25.68	24.81	25.68	25.68	28.57	30.08	28.57	28.57	29.67	17.28
		mean	27.39	24.93	27.38	25.33	27.01	26.75	27.61	29.13	29.77	28.74	27.44	18.62
		SD	9.45	5.14	9.13	5.44	8.77	10.07	8.97	9.48	11.39	9.35	8.62	8.29
		max	46.43	33.33	46.15	34.62	46.15	46.43	42.31	49.21	53.85	44.44	39.29	35.71
		min	11.11	12.82	11.11	12.82	11.11	11.11	14.74	17.05	17.05	15.77	13.33	7.69
	Class	median	78.8	75.94	78.8	76.69	77.88	25.56	41.03	42.31	42.31	41.03	42.31	40.14
		mean	73.63	72.92	73.36	73.18	73.74	27.26	42.16	43.97	44.21	42.76	44.83	41.69
		SD	9.53	9.21	9.57	9.22	9.64	5.28	13.44	13.81	13.55	14.37	13.67	12.56
		max	83.33	82.14	83.16	82.28	83.33	37.23	64.84	71.43	71.43	73.81	76.6	69.05
		min	57.69	57.14	56.41	57.69	57.69	21.58	22.08	26.14	25.61	21.75	25.79	22.11
RF	Statement	median	0	0	0	0	0	0	0	0	0	0	0	0
		mean	0.88	0.75	0	0.75	0	0.87	0.75	0.66	0.66	0.75	0.75	0.75
		SD	2.71	3.28	0	3.28	0	2.67	3.28	2.87	2.87	3.28	3.28	3.28
		max	10.24	14.29	0	14.29	0	10	14.29	12.5	12.5	14.29	14.29	14.29
		min	0	0	0	0	0	0	0	0	0	0	0	0
	Method	median	0	0	0	0	0	0	0	0	0	0	0	0
		mean	8.12	13.33	8.12	13.33	8.12	8.12	4.04	0.66	0.66	4.04	5.69	9.17
		SD	18.42	31.74	18.42	31.74	18.42	18.42	11.9	2.87	2.87	11.9	13.64	24.27
		max	66.67	100	66.67	100	66.67	66.67	50	12.5	12.5	50	50	100
		min	0	0	0	0	0	0	0	0	0	0	0	0
	Class	median	25	25	25	20	25	0	0	0	0	0	0	0
		mean	41.49	34.24	41.49	27.44	41.49	16.85	3.12	2.26	2.63	3.12	2.07	5.92
		SD	38.09	32.54	38.09	30.62	38.09	24.37	7.62	6.3	7.88	7.62	6.45	13.42
		max	100	100	100	100	100	66.67	25	25	25	25	25	50
		min	0	0	0	0	0	0	0	0	0	0	0	0

The dependent variable values are expressed as percentage.

4.3.2 Discussion. On the basis of the obtained results, it seems that *inadequate approaches achieve a better tradeoff between reduction in test suite size and loss in fault detection capability with respect to the adequate ones*. In particular, it seems that the best tradeoff is obtained by applying the instances of the CB approach (more specifically, CB_{EUCL}, CB_{COS}, CB_{JACC}, and CB_{HAMM}) when working at statement coverage level: high reduction of the original test suites with a small effect on the fault detection capability of these reductions (the difference is statistically significant as the results of our post-hoc analysis suggest).

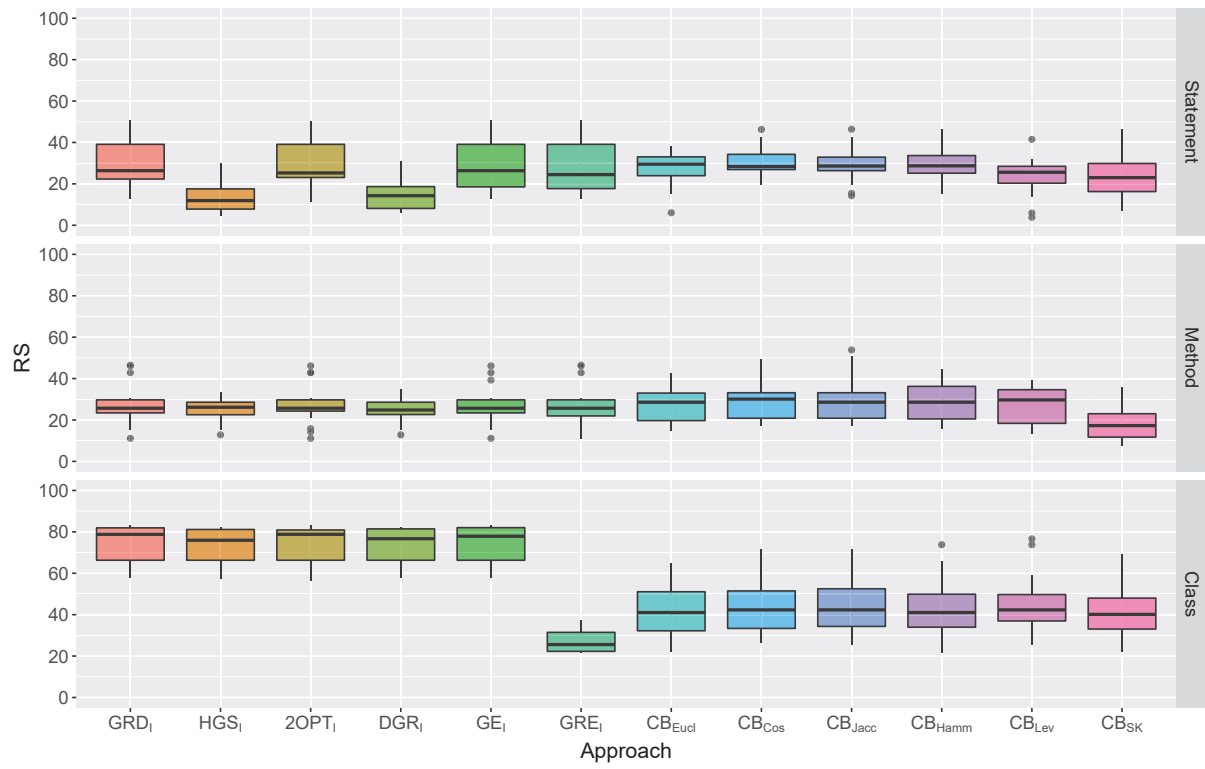
4.4 Implications and future extensions

We focus on the researcher and the practitioner (*i.e.*, tester) perspectives for the discussion of the implications and future extensions for our research.

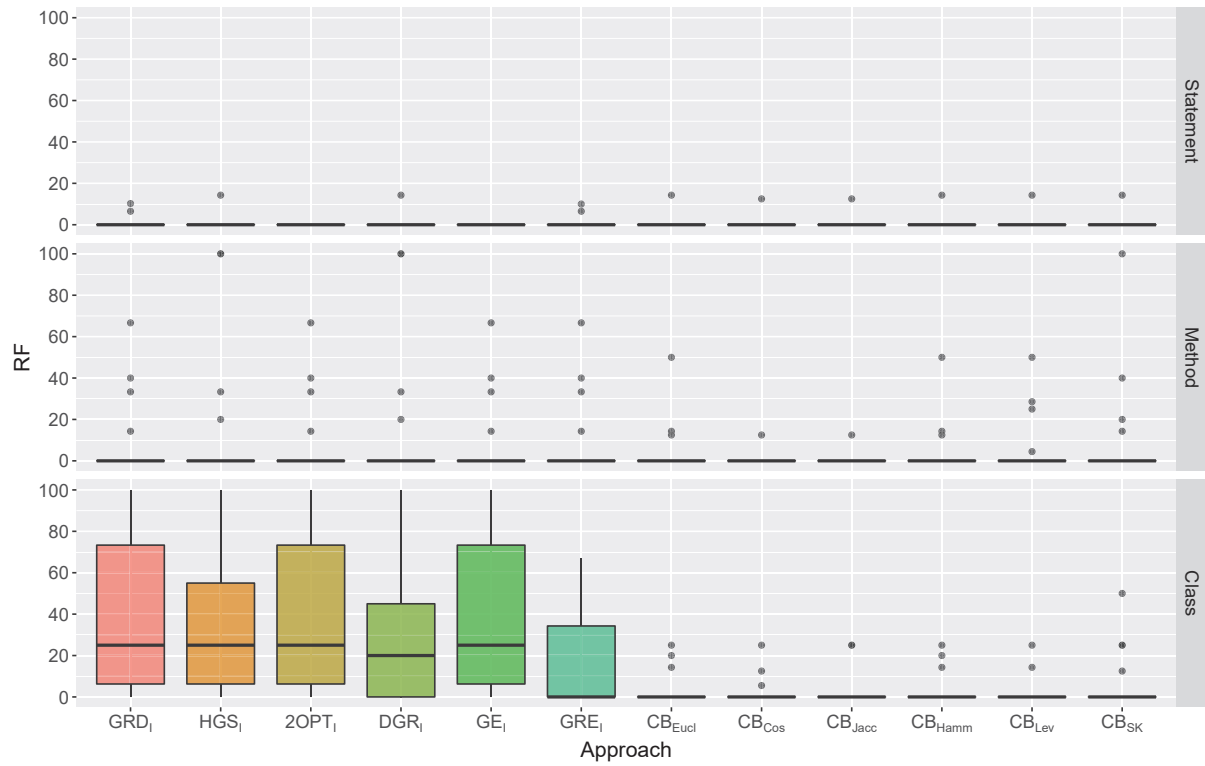
- Inadequate TSR approaches can be considered competitors of adequate ones although they relax constraints on test requirements. This is relevant for the researcher interested in further studying this point, but it is even more relevant for the practitioner. In fact, the practitioner can adopt inadequate

approaches being conscious that they will reduce a test suite more than adequate ones with a small effect on the capability of the reduced test suites to detect faults. 44

- Given a kind of coverage, adequate TSR approaches behave similarly. This is especially true for statement and method coverages. As for class coverage, GRE reduces less than the other approaches and its fault detection capability is higher. This result is relevant for the practitioner interested in non relaxing constraints on test requirements. That is, she could decide to use any of the adequate approaches, applied at either statement or method coverage level, being aware that the reduction in test suite size and the reduction in fault detection capability are nearly the same whatever the approach is. On the other hand, the researcher could be interested in further studying the effect of test requirements when dealing with adequate approaches and their effect on the tradeoff between the reduction in size and the reduction in fault detection capability of the reduced test suites. Our results might justify further researcher on this subject.



(a)



(b)

Figure 2: Boxplots for RS (a) and RF (b) for each inadequate approach and each kind of coverage.

- Inadequate approaches behave similarly, when applied at either statement or method coverage level, in terms of tradeoff between reduction in test suite size and loss in fault detection capability. Indeed, HGS_I and DGR_I applied at statement coverage level, and CB_{SK} applied at method coverage level, seem to perform slightly worse than the other approaches. These findings are relevant for the practitioner because she can apply any of the studied inadequate approaches at either statement or method coverage level being conscious of getting a good tradeoff in test suite size and loss in fault detection capability. The practitioner can decide to exploit the instances of the CB approach in case she is interested in reducing more the size of the original test suites with a small effect on fault detection capability.
- The use of class coverage allows (adequate and inadequate) TSR approaches to reduce more the size of the test suites. However, there could be a price to pay in terms of loss in fault detection capability of the reduced test suites. In particular, it seems that a coarse-grained information on code coverage has a detrimental effect on fault detection capability only on determined approaches (e.g., DGR_I). This implies that the researcher could be interested in studying to what extent and why some approaches are more or less sensitive to test requirements not studied in our experiment. Again, our results seems to justify further research on this subject.
- The use of the studied approaches does not require a complete and radical change process within a software company. This outcome can be considered relevant for the practitioner. In fact, the diffusion of a new technology/method is made easier when empirical evaluations are performed and their results show that such a technology/method solves actual issues [11]. This is why the results of our study could promote the transferring of the developed technology to the software industry. This is of particular interest for the practitioner, while the researcher could be interested in identifying opportunities (e.g., industrial case studies and experiments) to speed up this process.

5 RELATED WORK

Researchers have conducted a number of empirical studies on TSR. Chen and Lau [2] presented the results of a simulation study (*i.e.*, it relies on simulated data) of four adequate TSR approaches. The goal of the study was to define guidelines for choosing the most appropriate approaches for TSR. The definition of these guidelines was based on the size of the reduced test suites the approaches produced on simulated data. The most important result is that the studied approaches behave differently on the basis of the number of tests satisfying the same test requirements. Also Zhong *et al.* [25] based their study on the size of the reduced test suites. The authors focused on four adequate TSR approaches, which were applied on 11 C programs. The main result is that different TSR approaches can produce quite different reduced test suites although they had similar sizes. Both these studies (*i.e.*, [2, 25]) did not investigate the loss in fault detection capability of the reduced test suite. From this perspective, our study is complementary to those previously mentioned because we considered also the loss in fault detection

capability of the reduced test suites. Another difference with respect to these two investigations is that we also studied a larger number of TSR approaches (both adequate and inadequate) and considered larger and real software programs.

Wong *et al.* [21, 22] and Rothermel *et al.* [13, 14] assessed TSR with respect to both reduction in test suite size and loss in fault detection capability. Wong *et al.* [21, 22] studied an adequate TSR approach in the context of C programs. In particular, Wong *et al.* in [21] considered 10 small programs, while the authors studied a single large program in [22]. In both the papers, the authors observed a negligible loss in fault detection capability when reducing test suites. This result is coherent with that presented in this paper, *i.e.*, the loss in fault detection capability is negligible for adequate TSR approaches, when using either statement or method coverage. Rothermel *et al.* [13, 14] reported that losses in fault detection capabilities can increase when reducing test suites. Also this finding is confirmed in our study. In particular, when changing kind of test requirements (e.g., class coverage) the losses in fault detection capability could increase (or decrease). Unlike our empirical investigation, the ones above-mentioned ([13, 14, 21, 22]) studied only one adequate TSR approach, while we focused on a number of adequate and inadequate TSR approaches.

Zhang *et al.* [24] conducted an experiment where they evaluated four adequate TSR approaches on Java programs obtained from SIR. The authors took into account two kinds of test requirements (statement and method coverage). The results suggest that there is not a huge difference among the considered TSR approaches. This work is the most similar to ours, which might be considered an extension since we studied a further test requirement (*i.e.*, class coverage), more adequate (e.g., DGR) and inadequate (e.g., DGR_I) approaches. Our findings seem to confirm those by Zhang *et al.* and provide further empirical evidences in the context of TSR. Shi *et al.* [15] presented an empirical study on some adequate TSR approaches (e.g., HGS and GRD) and their inadequate versions to verify the benefits that can be achieved when relaxing the constraint of covering all the test requirements. The authors observed that for the adequate approaches, the reduction in test suite size is (median of) 62.9%, and by dropping down to 95% of the test requirements, such a reduction can increase 17.14%. Our study extends in some sense also that by Shi *et al.* [15] since we studied more TSR approaches (both adequate and inadequate) and other kinds of tests requirements, e.g., the authors studied neither method coverage nor class coverage. Our study confirms the results by Shi *et al.* [15] so increasing our body of knowledge about adequate and inadequate TSR approaches.

Smith and Kapfhammer [18] explored, through an experiment, the combination of two kinds of test requirements—statement coverage and test case execution time—to reduce test suites (and to prioritize test cases). In particular, they investigated four TSR approaches with three kinds of test requirements: statement coverage, test case execution time, and ratio of statement coverage to test case execution time. The results suggest the combining statement coverage with test case execution time allows achieving slightly better results in terms of reduction in test suite size. However, this study did not evaluate TSR approaches with respect to the loss in fault detection capability (as done in our study). This represents the most important difference with respect to our study.

Shi *et al.* [16] presented a study that compares TSR and regression test selection approaches. This comparison was based on statement coverage and killed mutants. The results on 15 open source projects suggested that regression test selection on average runs fewer tests than using TSR would. Shi *et al.* also found that a reduced test suite can result in a high loss in fault detection capability.

6 CONCLUSION

In this paper, we present an experiment to study adequate and inadequate TSR approaches in terms of tradeoff between reduction in test suite size and loss in fault detection capability. In particular, we studied six adequate TSR approaches and their inadequate variants. We also considered a novel (inadequate) clustering based approach and six instances of this approach. For each of these approaches, we considered three different kinds of test requirements: statement, method, and class coverages. All the approaches were applied on 19 Java programs from a public dataset widely used in the regression testing field, *i.e.*, SIR.

The most important result from our experiment is that inadequate approaches, as compared with adequate ones, allow obtaining the best tradeoff between reduction in test suite size and loss in fault detection capability. This is especially true when these approaches work with statement and method coverages. That is, the kind of test requirement investigated in our study can make the difference. For example, choosing class coverage as test requirement allows inadequate approaches to reduce more at the cost of a loss in fault detection capability. Indeed, the loss in fault detection capability seems acceptable in the case of the clustering based approach.

In addition to the future directions of our research delineated in Section 4.4, we plan to conduct users' studies (*e.g.*, case studies) to assess the effective benefit in using either adequate or inadequate TSR approaches. Investigating on how testers perceive these approaches and whether software industry is interested or not in their use represents another possible future direction.

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